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Project #19: Using Interprofessional Collaboration to Reduce Reported Rates of Central Line-Associated Bloodstream Infection in an Intensive Care Setting

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AIM

Central line-associated bloodstream infections (CLABSI) are preventable healthcare-associated infections (HAIs) associated with increased morbidity, mortality, and prolonged hospital stay.^{1,2} Additionally, CLABSIs are the most expensive HAIs with an estimated cost of \$46,000 per case.³

Problem Statement: The surgical intensive care unit (SICU) at Henry Ford Hospital experience high rates of CLABSI in 2019 and 2020.

Improvement Statement: By end 2022, the goal was to reduce the CLABSI rate per 1000 central line (CL) days for patients on the SICU by at least 50%.

PLAN

A planning group formed to reduce the occurrence of CLABSI on the SICU at Henry Ford Hospital.

- Review of surveillance criteria:** In an acute care setting, CLABSI surveillance criteria are based on the National Healthcare Safety Network (NHSN) definitions.⁴ Surveillance consists of review of positive blood cultures (BC) from patients with qualifying central lines (CL). Qualifying CL must be in place for > 2 consecutive calendar days after being accessed to meet criteria for CLABSI.⁵ A 7-day infection window period (IWP), which includes the 3 days before and after the first positive diagnostic test, is used to review infection criteria to meet the surveillance definition (Table 1). If no secondary sources are identified as a cause of the bloodstream infection (BSI), it will qualify as a CLABSI.⁵ Surveillance definitions are standardized to avoid variations in reporting. However, several factors, including misidentification of NHSN criteria and lack of diagnostic workup to identify secondary sources or documentation, could impact CLABSI reporting.
- Review of root cause analysis (RCA) data:** Multiple opportunities impacting unit CLABSI rates were identified, including gaps in clinical assessment, diagnostic testing, BC and CL utilization, and documentation. The team acknowledged that when an RCA was issued, it was already too late to identify infection source and avoid CLABSI. The interprofessional team decided to change workflow from reactive to proactive.

DO

February 2021: An interprofessional team formed, which included the unit medical director, infection prevention specialist, and clinical nurse specialist. The team communicated using secure email or texting group. Electronic medical records (EMRs) were reviewed of all patients with CLs and ordered BC. BC stewardship guidelines were followed by the interprofessional team, providing guidance on appropriate BC utilization. When BCs were not indicated, the care team was contacted for discontinuation. The EMR was also reviewed for infection source from existing diagnostic testing and documentation during the infection window period. Identified gaps were immediately communicated with patient care teams. Follow-up included collaborative efforts regarding proper documentation of clinical findings and treatment plan (Figure 1).

2nd Quarter 2021: Daily documentation of CL indications was implemented in the ICUs at Henry Ford Health. This was presented at Henry Ford Hospital CLABSI Team Meetings and System CLABSI Collaboration. The interprofessional team reinforced the assessment of CL and facilitated prompt removal of unnecessary lines.

Ongoing Education: Revised resident, fellow and RN orientation. Infection prevention in-services were provided. Cases that met secondary BSI were reported out during SICU collaborative, safety huddles, and other educational platforms to communicate and celebrate successes.

Table 1: Example of Infection Window Period

3 Days Before			Date of Event	3 Days After		
1-Jan	2-Jan	3-Jan	Blood Culture Collected	5-Jan	6-Jan	7-Jan

CHECK

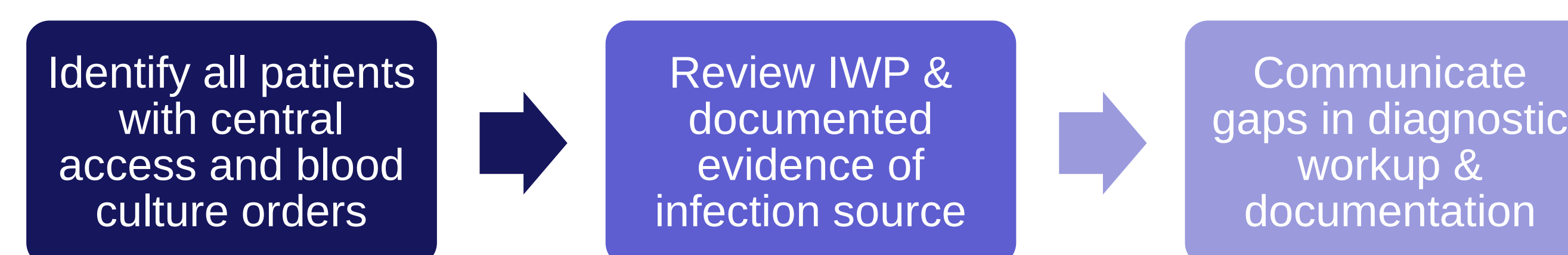
Review of EMR: Initially chart reviews were completed by the team twice per day, once in the morning and again in the evening. The chart review was found to be necessary, but the frequency was not. 2nd Quarter 2021: Frequency of chart reviews were decreased to once daily by a team member.

Consultation: There were multiple incidences when the team needed guidance on complex cases. 3rd Quarter 2021: The team partnered with the Infection Prevention Medical Director for assistance with complex cases.

Communication: 4th Quarter 2021: The team found the secure texting group to be the most useful and decided to use this for all communications moving forward. Email was eliminated as a method for communication for chart review report outs between the interprofessional team.

Additional Opportunities: The team identified miss opportunities to cancel pending BCs on end of life or expired patients. 3rd Quarter 2021: Updated the SICU Discharge Deceased Tipsheet to include BC cancellation process for qualifying patients. The process was reviewed and approved by the Henry Ford Hospital CLABSI Team Meeting and System CLABSI Collaboration.

Figure 1: Intervention Process



MEASURES

Data from the pre-intervention period (January 2019- January 2021) was compared to the intervention period (February 2021- December 2022).

- The interprofessional team identified secondary BSI in 37 patients (17 in 2021 and 20 in 2022) with CL and positive BC.
- CLABSI rate significantly decreased on the SICU from 1.78 in the pre-intervention period to 0.33 in the post-intervention period ($p < 0.001$), resulting in an 82% reduction. Similarly, the SIR declined from 1.469 to 0.279 ($p = 0.001$) with a resultant 81% reduction. Forty-six (3.3%) BCs were discontinued in the post-intervention period compared to 3 (0.1%) in the pre-intervention period. BC order rate was reduced by 33% ($p < 0.001$) and CL utilization ratio by 17% ($p < 0.001$) after the intervention was implemented. In addition, the overall unit BSI rate was 26% lower in the post-intervention period (Table 2).
- When comparing pre- and post-intervention periods there was an 82% reduction in CLABSI rates, reducing estimated healthcare cost associated with CLABSI by \$874,000.
- In the pre-intervention period, 48% of patients who developed CLABSI were male compared to 100% in the intervention period. Average age of patients who developed CLABSI was 64 in the pre-intervention period compared to 58 in the intervention period. Nearly one third of patients with CLABSI expired during the pre-intervention period. None of the patients with CLABSI expired during the intervention period.

ACT

- Consistent messaging and education were critical to improve understanding of NHSN rules, CLABSI definitions, diagnostic stewardship guidelines and CL indications.
- Successes and lessons learned shared monthly on SICU Collaborative meetings and shift huddles.
- In response to this project, a Henry Ford Hospital ICU RCA workgroup was formed to identify missed opportunities during the infection window period. This committee is attended by hospital administration, unit leadership, and infection prevention staff. Since implementation reduction in CLABSIs have been made, demonstrating that this project can be implemented in other practice areas.
- 4th Quarter 2023:** The interprofessional team collaborated with Information Technology to create a patient list column that displays BC result status. This column would assist team members with CLABSI surveillance. The development is ongoing.

Table 2: Pre- and Post-intervention Data

Metrics	Pre-intervention	Post-intervention	Reduction	P-value
CLABSI rate per 1000 CL days	1.78	0.33	82%	<0.001
SIR	1.469	0.279	81%	0.001
BC order rate per 1000 CL days	311	207	33%	<0.001
BSI rate per 1000 patient days	13.75	10.13	26%	<0.001
CL utilization ratio per 1000 patient days	570	471	17%	<0.001

KEYS TO SUCCESS

- Dedicated unit professionals allocated to quality improvement were vital for success.
- A key to success was shifting the culture from reactive to proactive. This was completed by timely communication of the infection window period, along with communication of potential gaps in diagnostic workup and documentation.

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